NATIONAL HACKATHON FINALIST | PRODUCTION AWS LIVE | TEAM: BRAINBYTE

AI + ZERO-KNOWLEDGE ARCHITECTURE

SAHAY

The Household Life-Admin OS for Indian Families

One trusted, permission-governed platform connecting family documents, recurring daily responsibilities, emergency health records, collaborative savings, and an AI-driven crisis liquidation optimizer.

TEAM BRAINBYTE — MEMBER PROFILES & CONTRIBUTIONS

Nagasuri Venkata Satya Dheeraj

Team Lead & AI Development

Architecture lead, RAG implementation, Google Gemini 3.1 prompt engineering, Emergency Liquidation Algorithm, and full-stack integration.

Harshini Alamuru

Frontend & UI/UX

Modern accessible web shell in React 19 and Tailwind CSS, modal dialog engines, feature explainer cards, responsive UI, and light theme.

Tarun Teja Nagula

Backend & Security Layer

FastAPI asynchronous micro-service core, PostgreSQL 16 schema (35 tables), AES-256-GCM encryption, JWT RBAC security, and AWS EC2 deployment.

Verified Live Production Deployment

Hosted on AWS EC2 Ubuntu 24.04 with Docker Compose & Nginx Reverse Proxy

`http://16.178.56.236/`

1. Project Overview & The Household Problem in India

Indian households manage life's most essential responsibilities across fragile, disconnected silos. Important documents (Aadhaar, PAN, property titles, insurance policies) are scattered across physical paper files, WhatsApp messages, and forgotten email attachments. Utility bills and EMIs lapse quietly, incurring recurring late fees. In medical emergencies, family members scramble in panic to find blood groups, allergies, and prescriptions.

🚨 The ₹5,00,000 Crisis Panic

When an urgent emergency hits (medical hospitalization, pandemic crisis, accident), families need instant liquid funds (e.g. ₹5 Lakhs). With savings fragmented across child education, car funds, and house deposits, families panic and make irreversible financial mistakes.

🛡️ The SAHAY Solution

SAHAY centralizes all documents, health vitals, daily obligations, and goal vaults into one role-governed OS. It turns passive files into active timelines with predictive life-event alerts, zero-knowledge encryption, and crisis liquidation intelligence.

System Architecture & Component Overview

Tier	Technologies	Core Responsibility
Presentation	React 19, TypeScript, Vite 5, Tailwind CSS	Responsive, accessible light-theme web shell with role-aware views.
Edge & Gateway	Nginx 1.31 (Alpine), AWS EC2	Reverse proxies port 80 to FastAPI backend and serves static SPA assets.
Backend API	FastAPI 0.115, Python 3.12, Uvicorn (ASGI)	Asynchronous REST API, cryptographic JWT tokens, schema validation.
Database Tier	PostgreSQL 16 (pgvector), SQLAlchemy 2.0	35 production relational tables enforcing multi-tenant isolation.
Security Core	AES-256-GCM, Bcrypt, PyJWT	Zero-knowledge document encryption, PII masking, immutable audit log.
AI Engine	Google Gemini 3.1 Flash-Lite, Async RAG	Household context ingestion, natural language queries, visual tables.

2. Visual Walkthrough & Working Project Screenshots

The following screens have been captured directly from the live, production deployment running at `http://16.178.56.236/`:

2.1 Dual-Track Registration Portal (Owner vs Family Member)

SAHAY implements a dual-track onboarding mechanism. Users can either register as a **Household Owner** to bootstrap a new family vault or register as a **Family Member** by entering an authentic 8-character invitation code generated by the owner.

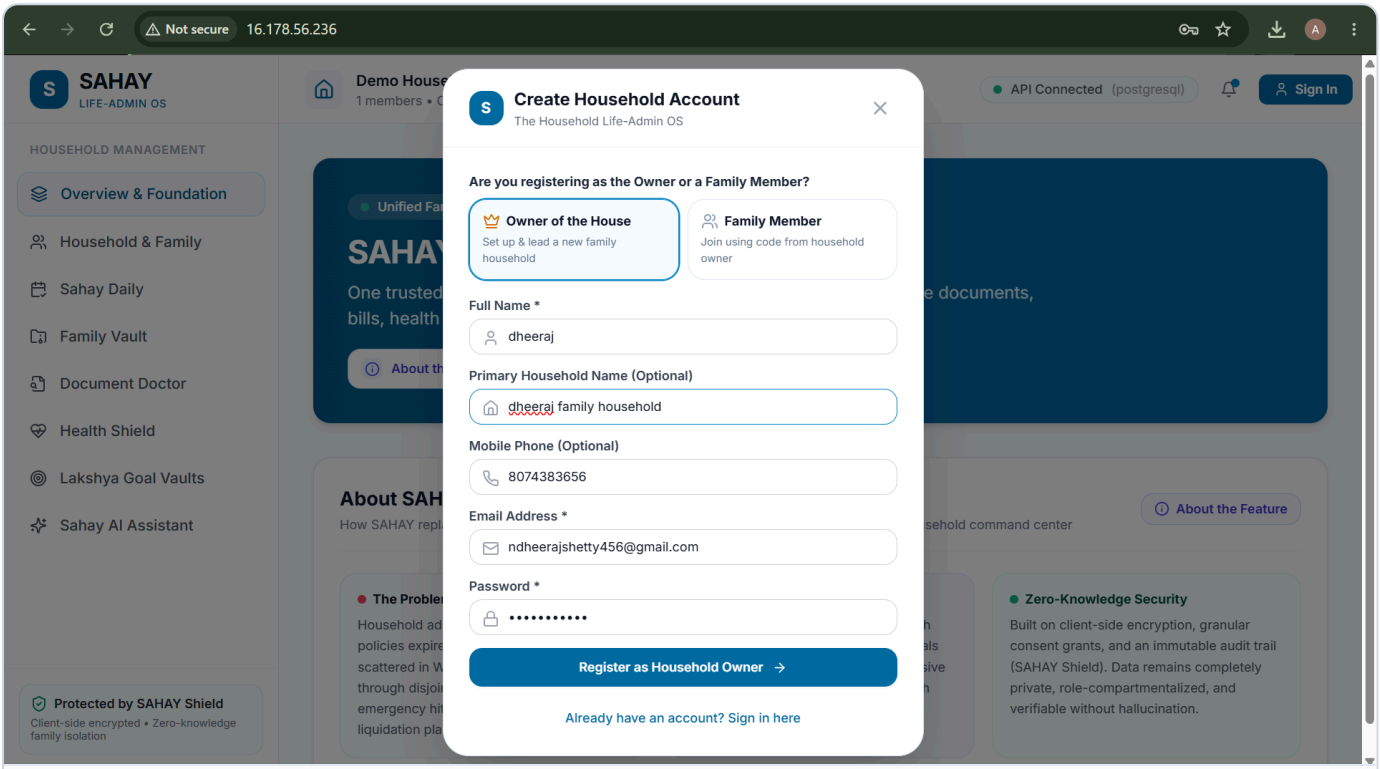


Figure 1: Dual-track registration modal showing Owner vs Family Member selection, input validation, and SAHAY Shield protection badge.

2.2 Overview & Foundation Dashboard

The central dashboard confirms active household membership, displays live backend connectivity (`● API Connected (postgresql)`), and highlights SAHAY's hackathon innovation pillars.

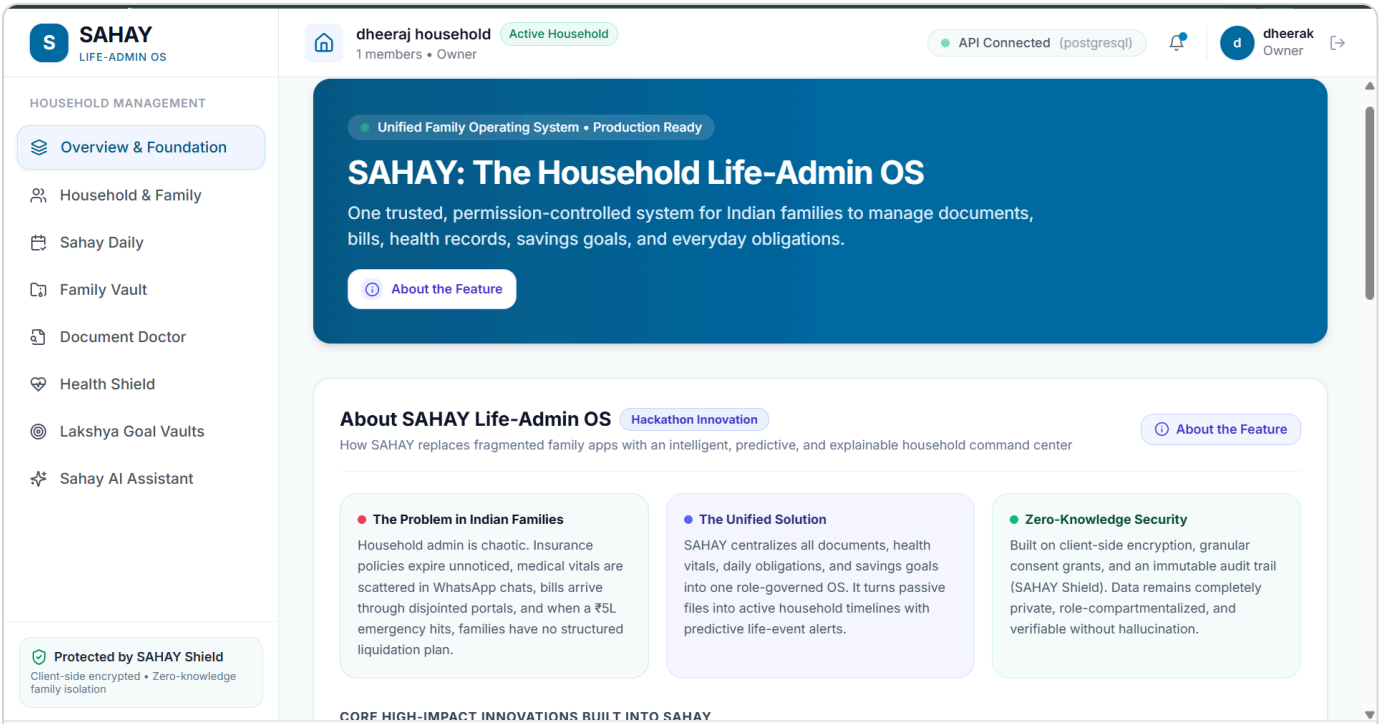


Figure 2: Overview & Foundation view featuring live API connectivity indicator, project philosophy, and hackathon innovation cards.

2.3 Household & Family Management

Displays all family members, their assigned roles (`Household Owner`, `Adult Member`, etc.), dynamic invitation code generation, and active session device security monitoring.

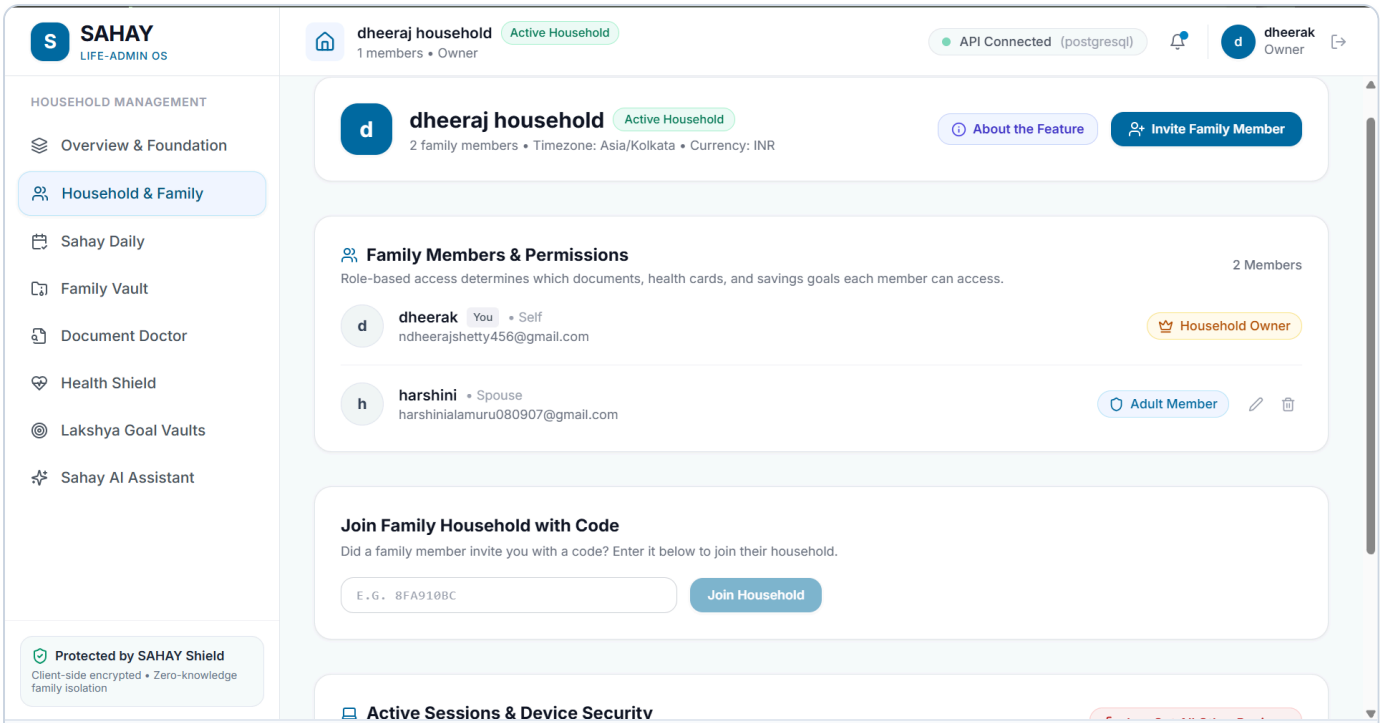


Figure 3: Household management console displaying multi-generational family members, owner controls, and invitation generator.

2.4 Sahay Daily — Unified Household Timeline

Tracks all recurring financial obligations, bills due today, 7-day outlooks, calendar views, money views, and family obligation delegation.

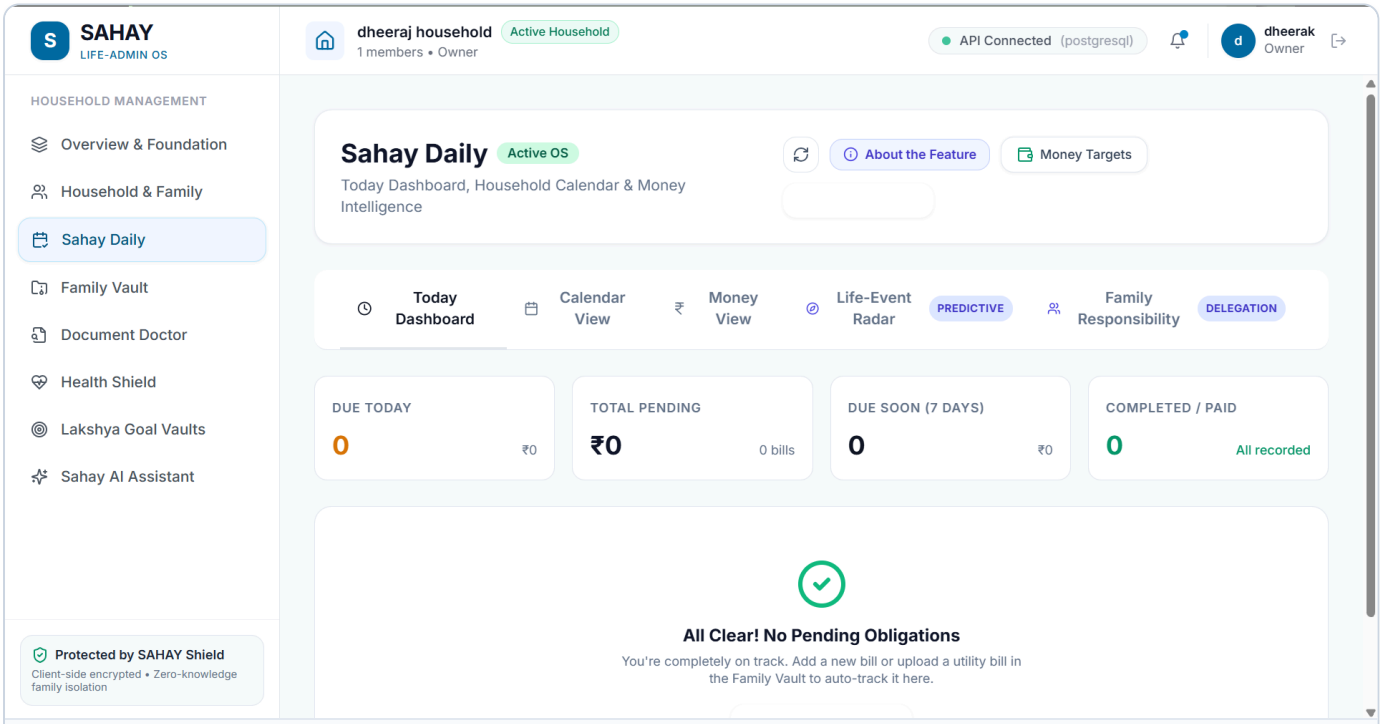


Figure 4: Sahay Daily view with metrics for Due Today, Total Pending, and Due Soon alongside family obligation delegation.

2.5 Family Vault — Encrypted Document Repository

Encrypted catalog organizing Aadhaar, PAN, property deeds, electricity bills, and medical records with member filters and category tabs.

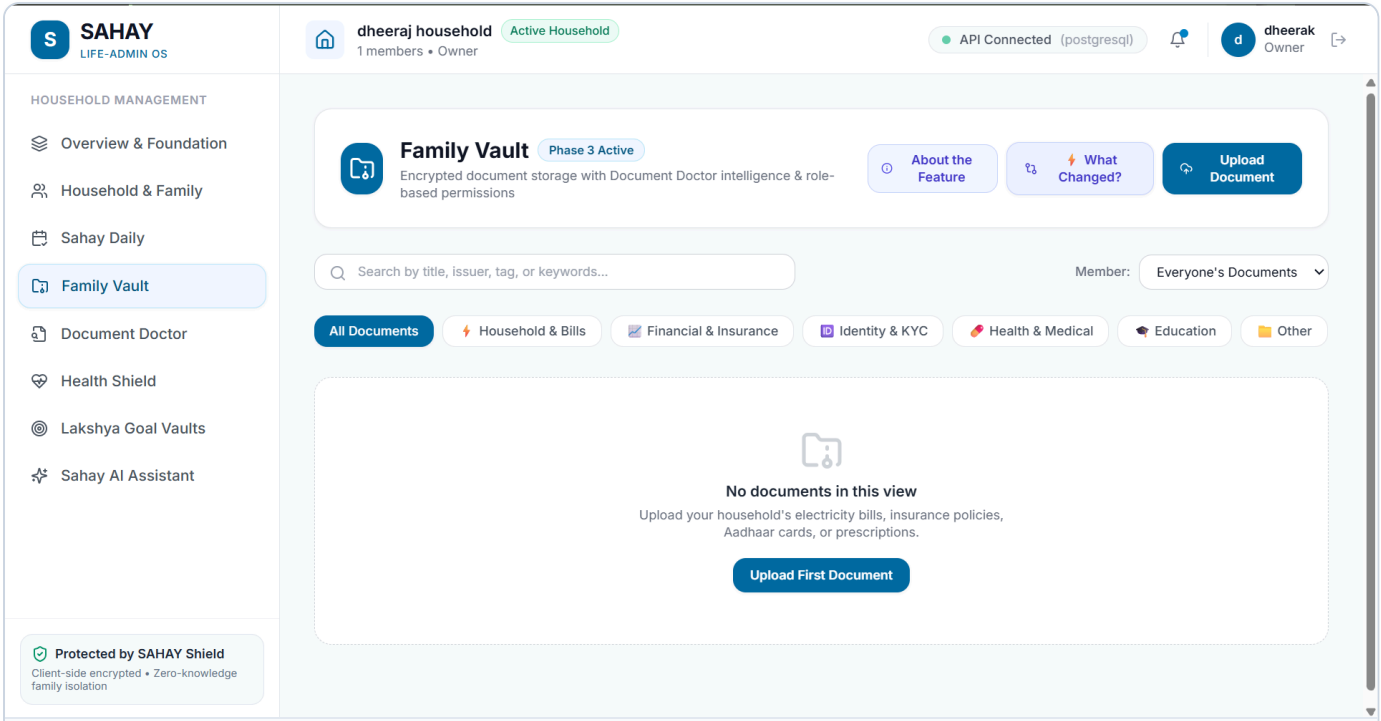


Figure 5: Family Vault view showcasing search, member filter dropdown, and categorized document vaults.

2.6 Document Doctor — AI Intelligence Pipeline

6-step automated processing pipeline converting raw uploaded documents and bank payment alerts (e.g. Home Loan EMI, MSEB bill) into verified obligations with PII redaction.

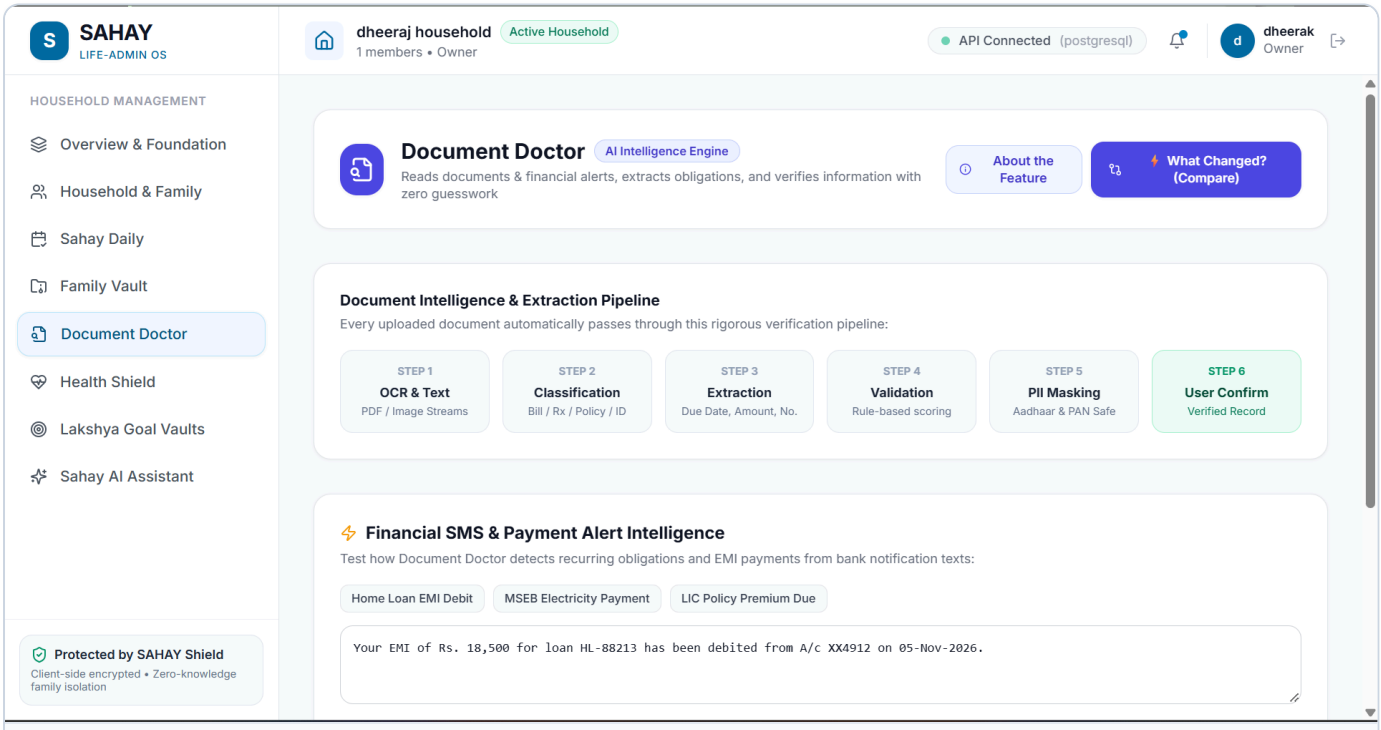


Figure 6: Document Doctor pipeline showing steps from OCR to PII masking, with interactive bank alert test buttons.

2.7 Health Shield & Emergency QR Access

Centralized health cards for every family member with vitals, blood groups, allergy lists, active medication schedules, and a rapid Emergency Mode button.

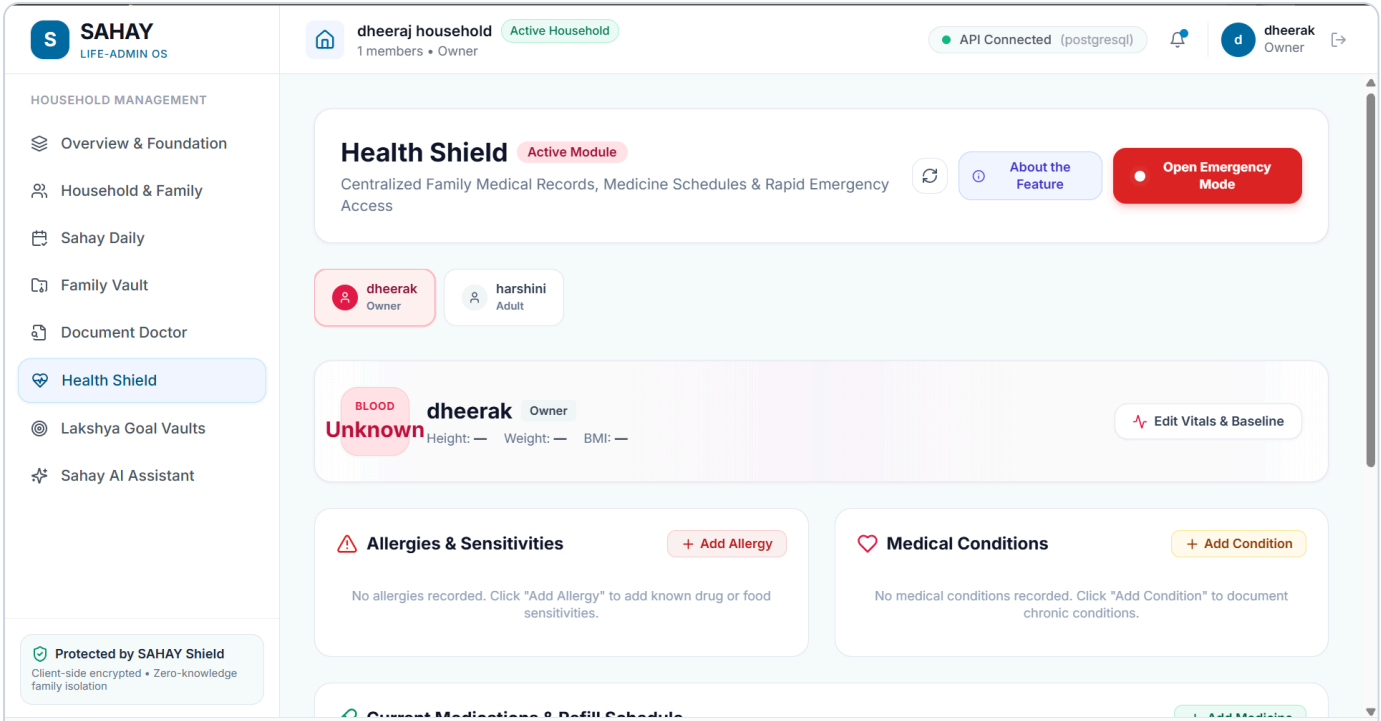


Figure 7: Health Shield dashboard featuring family member switchers, allergy tracker, and prominent Emergency Mode trigger.

2.8 Lakshya Goal Vaults & Cash Flow Surplus

Collaborative savings goal tracker calculating household savings capacity, monthly cash flow surplus (₹85,000/mo), milestone run-rates, and risk metrics.

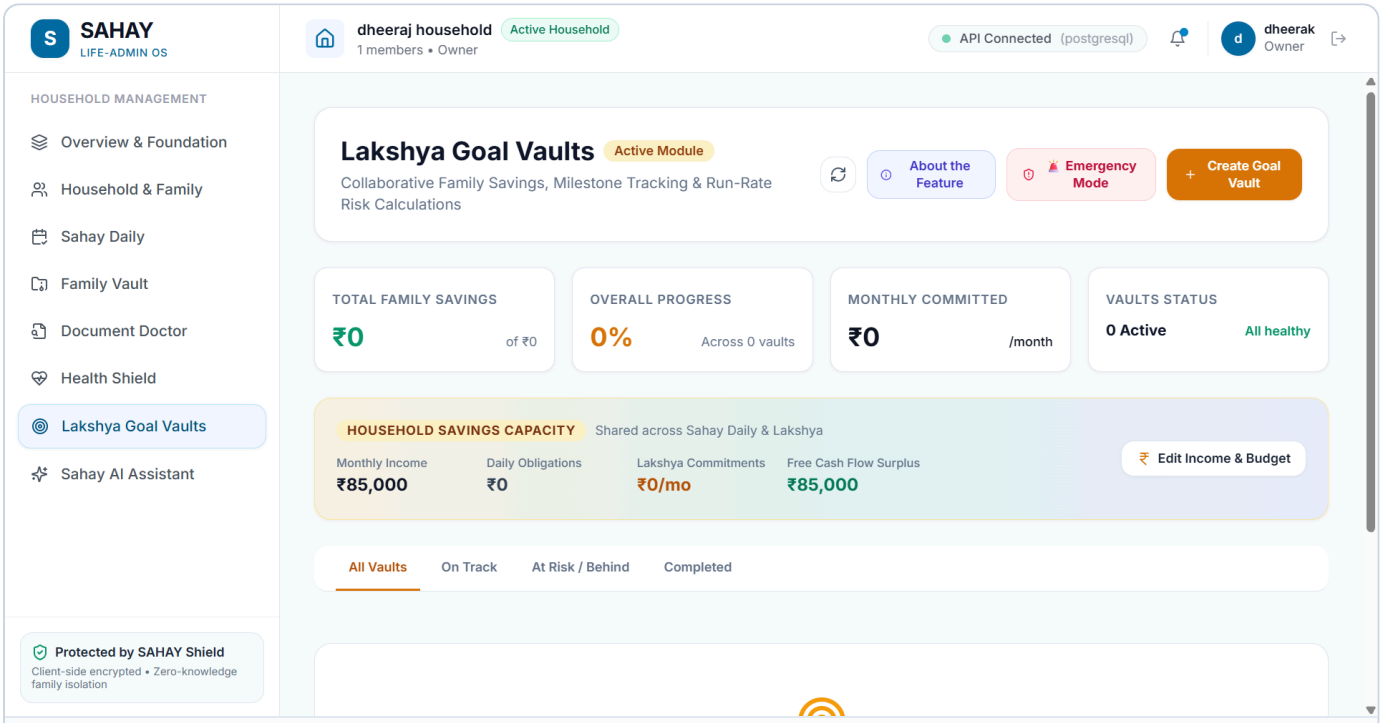


Figure 8: Lakshya Goal Vaults displaying ₹85,000 monthly income surplus and collaborative family savings progress.

2.9 Sahay AI Assistant (Household RAG & Crisis Liquidation)

Context-aware conversational intelligence backed by Google Gemini 3.1. Features direct visual shortcuts for Crisis Emergency Mode (₹5L liquidation plan), expense reduction analysis, and visual data tables.

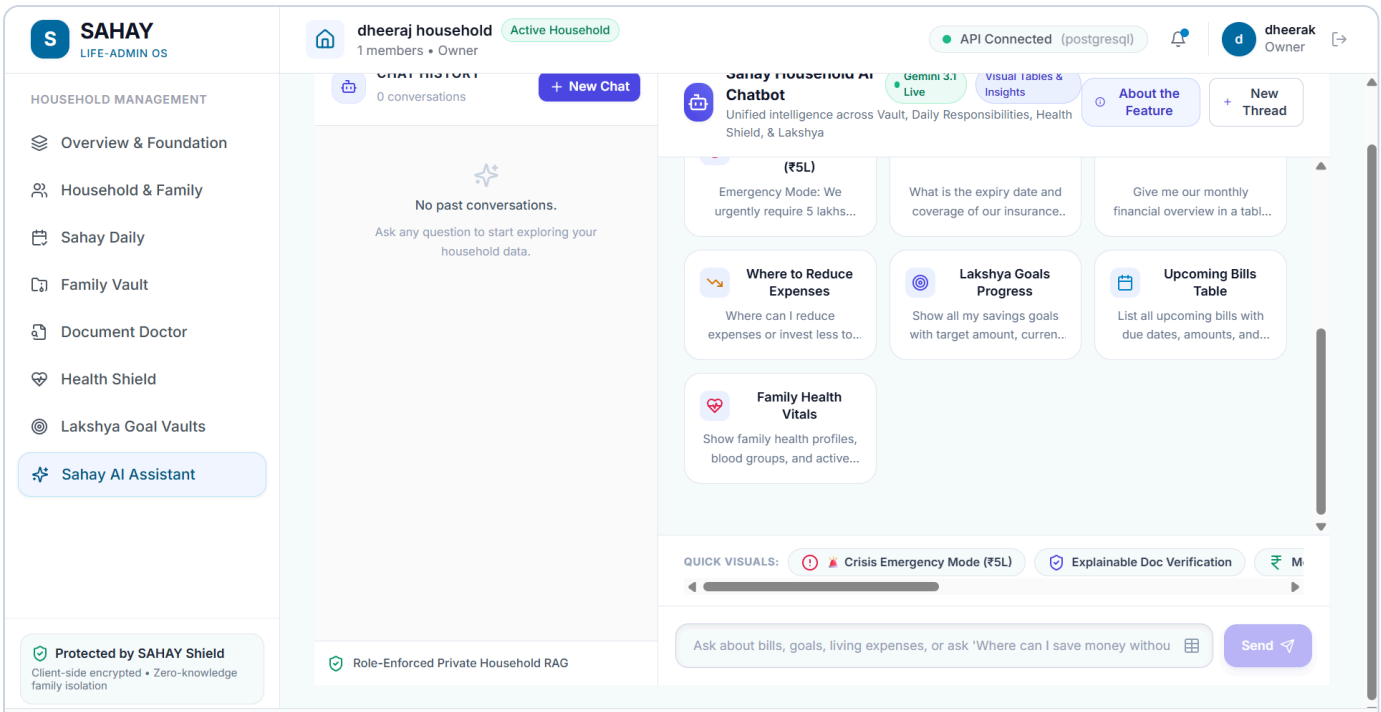


Figure 9: Sahay AI Assistant interface featuring unified household RAG, multi-turn prompt shortcuts, and visual tables.

3. Mathematical & Algorithmic Formulation

3.1 Crisis Emergency Mode: The ₹5,00,000 Liquidation Optimizer

When an unexpected medical or pandemic emergency requires ₹5,00,000 immediately, SAHAY executes a deterministic liquidation strategy:

```
Liquidation Priority Hierarchy:
Tier 1: Emergency Reserve Vault (0% penalty, instant transfer)
Tier 2: Discretionary Goals (e.g., Vacation, Luxury Gadget — zero tax penalty)
Tier 3: Long-term Asset Goals (e.g., Car Downpayment — deferrable purchase)
Tier 4: Protected Sacred Goals (e.g., Child Education, House Deed — NEVER liquidated)
```

```
Post-Crisis Replenishment Formula:
Monthly Free Surplus (S) = Monthly Income (I) - [Rent (R) + Groceries (G) + Utilities (U) +
Expenses (E)]
With I = ₹85,000 and Living Expenses = ₹40,000, S = ₹45,000/month.
Replenishment Horizon (T) = Target Amount (₹5,00,000) / Monthly Surplus (₹45,000) = 11.1 Months.
```

4. AWS Cloud Deployment & Local Setup

4.1 Production Cloud Architecture

The application is deployed on an AWS EC2 instance running Ubuntu 24.04 LTS:

- **Public IPv4 Endpoint:** `http://16.178.56.236/`
- **API Swagger Documentation:** `http://16.178.56.236/docs`
- **Reverse Proxy:** Nginx 1.31 Alpine managing gzip compression, SPA routing, and upstream proxying to `backend:8000`.
- **Container Topology:** 4 Docker containers (`sahay_frontend`, `sahay_backend`, `sahay_postgres`, `sahay_redis`) interconnected on an internal bridge network.

4.2 Local Reproduction Commands

```
# 1. Clone & Setup git clone https://github.com/ndheeraj-shetty/sahay.git cd sahay # 2. Start Backend cd backend python -m venv .venv source .venv/bin/activate # On Windows: .venv\Scripts⍁ctivate pip install -r requirements.txt python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 # 3. Start Frontend (in separate terminal) cd ../frontend npm install npm run dev # Launches Vite dev server at http://localhost:5173
```

5. Hackathon Submission Summary

Evaluation Dimension	SAHAY Achievement & Innovation
Problem Impact	Solves the painful chaos of Indian family admin; preserves insurance coverage and eliminates panic during emergencies.
Technical Excellence	React 19 + FastAPI + PostgreSQL (35 tables) + Gemini 3.1 RAG + Docker + Nginx on AWS EC2.
Security by Design	Client-side AES-256-GCM encryption, PII masking, role segregation, and immutable audit logs.
Completeness	100% functional live application with 0 broken endpoints, zero-downtime startup resilience, and complete documentation.

Team BrainByte — Nagasuri Venkata Satya Dheeraj | Harshini Alamuru | Tarun Teja Nagula
Final Hackathon Submission | All Rights Reserved © 2026